

DEPARTMENT OF TEXTILE TECHNOLOGY

CURRICULUM OF M. Tech. (Textile Technology) (With effective from 2023-2024)

First Year					
Course Code	Name of the course	L	T	P	Credits
Semester I					
PCC-TT-501	Advanced Yarn Production	3	0	2	4
PCC-TT-502	Melt Spinning and Texturizing	3	0	2	4
PCC-TT-503	Computer Applications in Textile	3	0	2	4
PEC-TT-5**	Program Elective – I	3	0	0	3
PEC-TT-5**	Program Elective – II	3	0	0	3
MCC-590	Research Methodology and IPR	2	--	--	2
MCC-591	English for Research Paper Writing	2	--	--	2
Sub-Total		19	0	6	22
Semester II					
PCC-TT-510	Modern Weaving Technology	3	0	2	4
PCC-TT-511	Evaluation Techniques of Textiles	3	0	2	4
PEC-TT-5**	Program Elective – III	3	0	2	4
PEC-TT-5**	Program Elective – IV	3	0	2	4
PRJ-TT-518	Mini Project and Seminar	0	0	2	1
OEC-8**	Open Elective	3	--	--	3
AUD-9**	Audit Course	2	--	--	--
Sub-Total		17	0	10	20
Second Year					
Semester I					
DIS-TT-601	Dissertation – I	0	0	36	18
HSSM-601	Indian Knowledge System	2	--	--	02
Sub-Total		2	0	36	20
Semester II					
DIS-TT-602	Dissertation – II	0	0	36	18
CCC-602	Yoga and Sports	--	--	02	--
Sub-Total		0	0	38	18
Grand Total		38	0	90	80

L – No. of Lecture Hours/week, T – No. of Tutorial Hours/week, P – No. of Practical Hours/week

for PAKHLE

[Signature]

Program Elective – I	
Course Code	Name of the course
PEC-TT-504	Smart Textiles
PEC-TT-505	Fiber Characterization & Mechanical Properties
PEC-TT-506	High Performance Fibers
Program Elective – II	
Course Code	Name of the course
PEC-TT-507	Textile Project Planning & Implementation
PEC-TT-508	Industrial Engineering & Management
PEC-TT-509	Industrial Law and Economics
Program Elective – III	
Course Code	Name of the course
PEC-TT-512	Technical Textiles – I
PEC-TT-513	Advanced Textile Wet Processing
PEC-TT-514	Garment Technology & Merchandising
Program Elective – IV	
Course Code	Name of the course
PEC-TT-515	Knitting & Nonwoven Technology
PEC-TT-516	Textile Composites
PEC-TT-517	Technical Textiles – II

Open Elective Courses	
Course Code	Name of the course
OEC-801	Business Analytics
OEC-802	Industrial Safety
OEC-803	Operations Research
OEC-804	Cost Management of Engineering Projects
OEC-805	Composite Materials
OEC-806	Waste to Energy
OEC-807	Cyber Security

Audit Courses	
Course Code	Name of the course
AUD-901	Project Management
AUD-902	Disaster Management
AUD-903	Sanskrit for Technical Knowledge
AUD-904	Value Education
AUD-905	Constitution of India
AUD-906	Pedagogy Studies
AUD-907	Stress Management by Yoga
AUD-908	Personality Development through Life Enlightenment Skills

for J.A.K.K.

**First Year M. Tech
SEMESTER-I**

Code	Name of the Course Contents	Total Credits	Lectures /Week	Practical/ Week
PCC-TT-501	Advanced Yarn Production	04	03	02
Course Contents Objectives: 1. To explain opening and cleaning aspects of modern blow room machinery and process parameters involved in it. 2. To describe design aspects of different zones of modern card to process parameters involved in it. 3. To describe design aspects of various components of a comber & theories of drafting. 4. To describe fibre blending and process parameters involved in it. 5. To describe advance aspects of various spinning technologies.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Critically analyze various aspects of opening and cleaning in modern blow room machinery and also process parameters involved in it 2. Critically analyze design aspects of different zones of modern card and process parameters involved in it. 3. Critically analyze design aspects of various components of a comber & theories of drafting. 4. Critically analyze fibre blending and process parameters involved in it. 5. Critically study advance aspects of various spinning technologies.				
Course Contents:- Unit I- Spinning Preparations: Fibre mixing and blending practices in industry, Bale Management, Blending Performance Evaluation – Degree of Mixing, Index of Blend Irregularity, Hamburger model-yarn tenacity prediction from fiber properties, Evaluation of blow room performances and AFIS applications Unit II - Fibre separation in carding, Design and actions of wire points in carding, Transfer efficiency in carding, Roller drafting, Irregularity in textile strands, Limit irregularity, Index of irregularity, Causes of generation of irregularities, Hook removal in draw frame, Auto levelers in carding/ draw frame, Details of combing preparation, Process parameters in speed frame and roving quality Unit III – Ring frame: Twisting and winding process in ring spinning, Yarn tension in spinning balloon and its effect on yarn quality, Mass variation in ring spun yarns- basics, causes, measurements and detections Unit IV- Rotor Spinning: Productivity- Charka, Ring and Rotor spinning, Fibre properties and yarn quality, Yarn tension in rotor spinning, Process and machine parameters and yarn properties Unit V - Air-Jet Spinning: Twisting and yarn formation, Process and machine parameters and yarn properties, Yarn structure and properties Unit VI - Friction Spinning: Mechanism of yarn formation, Production of multi-layer and multi-component yarns by friction spinning, Yarn structure and property relation				
References Books: 1. <i>Spinning of Manmade and Blends on Cotton System</i>, K. R. Salhotra 2. <i>Textile Mathematics, Vol.-2 (The Textile Institute)</i>, J. E. Booth 3. <i>The Technology of Short Staple Spinning, (Short Staple Spinning Series, Vol.-I)</i>, W. Klein 4. <i>A Practical Guide to Opening and Carding, (Short Staple Spinning Series, Vol.-II)</i>, W. Klein 5. <i>A Practical Guide to Combing and Drawing, Vol.-3 (The Textile Institute)</i>, W. Klein 6. <i>A Practical Guide to Ring Spinning, Vol.-4 (The Textile Institute)</i>, W. Klein				

for Spinning

7. *New Spinning Systems, Vol.-5 (The Textile Institute), W. Klein*
8. *Fundamentals of Spun Yarn Technology (CRC Press), Carl A. Lawrence*
9. *The Principles and Theory of Ring Spinning, Vol.-V (The Textile Institute), A. E. De Barr and H. Catling.*
10. *Open-end Spinning (Elsevier Science), V. Rohlena*
11. *Rotor Spinning- Technical and Economic Aspects (Textile Trade Press, The Textile Institute), E. Dyson*
12. *Rotor Spinning (The Textile Institute, Manchester), C. A. Lawrence and K. Z. Chen*

Practical:

Practicals are to be conducted based on the above Course Contents.

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PCC-TT-502	Melt Spinning and Texturizing	04	03	02
Course Contents Objectives 1. To describe the requirements of Melt spinning process 2. To explain the details of spinneret design 3. To illustrate the different parameters needed and raw material properties in spinning process 4. To understand the texturizing processes and different parameters needed therein.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Describe the significance and scope of synthetic fibres in textiles 2. Explain the technical details spinneret design. 3. Complete understanding the properties needed for synthetic yarn manufacturing 4. Complete understanding the manufacturing of textured yarn				
Course Contents:- Unit I - Introduction , melt spinning concepts of polyester, polypropylene, Nylon, spinning of LOY, POY, FDY, BCF, Flow chart & various components of a melt spinning line, functions and design of the extruder, manifold, Unit II - dynamic mixer, spinning beam, metering pump, melt filters, spinning pack, spinneret, types, design & role of quench chamber, spin finish applicator, winder, dryer, spin finish pump, intermingling device, Drawing & heat setting. Unit III - Raw material, types of chips, storage & conveying of chips, Chips properties, dew point, melt spinning line parameters & their controls, quench air parameters, winding parameters & calculations, testing & properties of raw materials & filament yarn, packaging of spools, norms for gradation, Unit IV -problems & remedial action during process, burn-out section, spin finish oil preparation, auto control systems, oil heating systems, Air treatment plant Unit V - Procedure for starting up of line, mounting & removal of spinning pumps, spin packs, Post spinning processes: texturizing, types, basic principles of draw & air jet texturising, yarn processing on texturising machine, processing parameters, yarn properties, problems & remedial actions during texturising, Unit VI -comparative study of texturising machines, modern developments, texturising for fancy yarn production. Modern developments in twisting & rewinding.				
Reference books: 1. <i>Manufactured fibre technology</i>, V B Gupta & V K Kothari, Chapman & Hall 2. http://nptel.ac.in/Course Contentss/ 3. <i>Winter School on Manmade Fibres Vol-I</i>; Edited By Gupta & Kothari; 1988,IITD 4. <i>Winter School on Manmade Fibres Vol-II</i>; Edited By Gupta & Kothari;1988,IITD Man-made fibres - Moncrieff. 5. <i>Production of Synthetic Fibers</i> by A. A. Vaidya, Fst Edition, published by Prentice Hall of India, New Delhi, 1988 6. <i>Yarn Texturing Technology</i> by J. W. S. Hearle, L. Hollick and D. K. Wilson; Textile Institute, WoodHead Publishing House, U. K. 7. <i>A Guide to Crimping/Texturising Technology</i> by MANTRA, Surat				
Practical: Practicals are to be conducted based on the above Course Contents.				

for *PA Khule*

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PCC-TT-503	Computer Application in Textiles	04	03	02
Course Contents Objectives 1. To describe the Object-oriented Programming using C++. 2. To explain the Relational Databases & E Commerce. 3. To illustrate the applications of ERP and its related technologies with real life examples. 4. To explain the applications of SAP & its advantages.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Describe the significance and scope of Programming using C++ 2. Explain the technical details of Relational Databases & E Commerce 3. Compile the various properties, merits and applications of ERP. 4. Evaluate the suitability of SAP for various applications				
Course Contents Unit I- Object-oriented Programming using C++: Introduction to object oriented programming, basic program construction, variable types, loops & decisions, structures, functions, objects & classes, arrays, polymorphism, operator overloading, function overloading, inheritance Unit II - Relational Databases: Relational Model, Database Users, Roles of Database Administrator, keys, Domain Constraints, Referential Integrity, Structured Query Language (SQL), Database recovery methods Unit III - E-Commerce: The scope of electronic commerce, definition of electronic commerce-e-commerce and the trade cycle, Electronic markets, Electronic data interchange, Internet Commerce, Business Strategy in E-commerce. The value chain, supply chain, Porter's value chain model. Inter organization value chains, Business to business E-commerce, Inter organizational transaction, the credit transaction trade cycle. Advantages & disadvantages of Electronic markets. Application of E-commerce in textile industries. Unit IV -ERP and Its Related Technologies: Introduction to ERP, Basic ERP concepts, Justifying ERP Investments, RISK of ERP, Benefits of ERP. ERP and Related Technologies, Business Process Re-engineering (BPR), Product Life Cycle Management, Supply Chain Management (SCM), Customer Relationship Management (CRM). Use of ERP in Textile Industry. Unit V - SAP: Architecture of SAP R/3, SAP Integrated- Analysis, Implementation and Design, Three-Tier Architecture, Need of Multi-tier Architecture, Integrating Environments. Unit VI - Business Intelligence System: Technical Architecture overview, Back room Architecture, Presentation Server Architecture, Front room Architecture, Metadata, Standard Reports, Dashboards and Scorecards				
Reference Books 1. Object Oriented Programming with C++ -E. Balagurusamy. 2. Database System Concept by Henry F. Korth, Abraham Silberschatz, Sudarshan (McGraw Hill Inc.) 3. E-Commerce --David Whiteley, TmH. 4. ERP Demystified -Alexis Leon, TMH 5. Enterprise Resource Planning --Alexis Leon, TMH. 6. SAP R/3 SAP Architecture, Administration, Basis, ABAP Programming with MM and SD Modules – Dreamtech Press. 7. The Data Warehouse Lifecycle Toolkit By Ralph Kimball, Ross, 2nd edition, Wiley Publication				
Practical: Practical's are to be conducted based on the above Course Contents				

List of Course Contents for Program Elective-I – (Any One)

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-504	Program Elective-I - Smart Textiles	03	03	00
Course Contents objectives: 1. To describe general introduction of smart textiles and modeling of intelligent materials. 2. To elucidate temperature sensitive shape memory polymers. 3. To explain solar textiles and introduction to conductive materials. 4. To understand applications of smart / intelligent textiles.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Remember general introduction of smart textiles and modeling of intelligent materials. 2. Understand solar textiles and conductive materials. 3. Analyze temperature sensitive shape memory polymers. 4. Interpret applications of smart / intelligent textiles.				
Course Contents Unit I – General introduction: Definition, classification, intelligent systems and general applications. Unit II - Modeling of intelligent materials: Background, underpinnings of interdisciplinary, scientific practices and research strategies for intelligent garments Unit III – Phase change materials: Heat balance and thermo-physiological comfort, Phase change technology, PCM in textiles, Future prospects of PCM in textiles and clothing Intelligent textiles with PCMs: Basic information of phase change materials, Phase change properties of linear alkyl hydrocarbons. Unit IV – Temperature sensitive shape memory polymers: A concept of smart materials, Shape memory polymer and smart materials, Some examples of shape memory polymer for textile applications, Potential use of shape memory polymer in smart textile, General field of application, Challenges and opportunities. Study of shape memory polymer films for breathable textiles: Breathability and clothing comfort, Breathable fabrics, Water vapor permeability (WVP) through shape memory polyurethane Chromic and conductive materials: Photo chromic materials, Thermo chromic materials, Color changing, Electro chromic materials Unit V – Solar textiles: production and distribution of electricity coming from solar radiation: Solar cells, Textiles as substrates, Technological specifications, Challenges to be met, Suitable textile constructions. Introduction to conductive materials: Electric conductivity, Metal conductors, Ionic conductors, inherently conducting polymers, Application technologies for conducting fibre materials Multipurpose textile based sensors: Introduction, Conductive polymer textile sensors, Conductive polymer composites (CPCs) textile sensors Textile micro system technology: Textile micro system technology, Textiles are inherent microstructures, Textile-based compliant mechanisms in micro-engineering and mechatronics Unit VI - Applications: Intelligent textiles for medical and monitoring applications, Context aware textiles for wearable health assistants, Intelligent garments in prehospital emergency care, Intelligent textiles for children, Wearable biofeedback systems, Applications for woven electrical fabrics				
Reference Books 1. Smartfibres, fabrics and clothing edited by Xiaoping Tao, Woodhead publishing Ltd., England. 2. Intelligent Textile and clothing edited by H.R.Mattila, Woodhead Publishing, England. 3. Clothing bisensory Engineering edited by Y.L. and A.S.W Wang, Woodhead publishing ltd. England.				

for package

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/Week
PEC-TT-505	Program Elective-I Fiber Characterization & Mechanical Properties	03	03	00
Course Contents Objectives 1. To describe the fibre structure and morphology 2. To explain the Tensile properties of fibres 3. To illustrate the Theories of mechanical properties with numerical examples 4. To evaluate the structural parameters by investigation techniques.				
Course Contents Outcomes At the end of the Course Contents students will be able to 1. Understand the fibre structure and morphology 2. Analysis the technical details of Tensile properties of fibres 3. Illustrate the theories of mechanical properties with numerical examples 4. Analysis and evaluation of fiber structure by different methods & Techniques.				
Course Contents Unit I - Mechanism of deformation in fibers, Principles of elasticity & Visco elasticity, Unit II - Stress-strain relations, Creep, Stress-relaxation, Time-temperature effects, Unit III - Dynamic mechanical properties, Model theory of visco-elasticity, Unit IV - Thermodynamic analysis of mechanical deformation & rubber elasticity, Unit V - fibre friction optical properties of fibers, Unit VI - Refractive index & birefringence. X-ray diffraction, Electron microscopy, IR spectroscopy, Scanning electron microscopy & study of fine structure, Surface structure of different textile fibres.				

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-506	Program Elective-I High Performance Fibers	03	03	00
Course Contents Objectives 1. To describe the Requirements of high performance (HP) fibres 2. To explain the Manufacturing of aramids, carbon, glass and chemical resistant fibres 3. To illustrate the properties of aramids, carbon, glass and chemical resistant fibres 4. To explain the applications of aramids, carbon, glass and chemical resistant fibres				
Course Contents Outcomes At the end of the Course Contents students will be able to 1. Describe the significance and scope of HP fibres in technical textiles 2. Explain the technical details of Glass, Aramids, Carbon and other HP fibres 3. Compile the various properties and merits of above stated fibres in technical textiles 4. Evaluate the criteria for applications in technical textiles and their cost.				
Course Contents Unit I - Significance of High-Performance Fibres. Critical Comparison of Regular and High-Performance Fibres, Review of Various Fibre Manufacturing Processes. Unit II - Manufacturing of Aramid Fibres, Analysis of Structure and Characteristics of Important Aramid Fibres, Comparison of Characteristics of Important Commercially Available Aramid Fibres, Studies on the Applications of Aramid Fibres. Unit III - Manufacturing of High-Performance Polyethylene and Fully Aromatic Polyester Fibres, Analyses of Characteristics of High-Performance Polyethylene Fibres and Fully Aromatic Polyester Fibres. Unit IV - Studies on the Applications of These Fibres. Inorganic High-Performance Fibres: Glass Fibre Manufacture, Properties, and Applications. Ceramic Fibres: Analysis of Characteristics and Applications of Silicon Carbide-based Fibres, Alumina-based Fibres, Single Crystal Oxide Fibres. Unit V - Critical Analyses of Fibre Characteristics and Applications of Chlorinated Fibres: PVDC. Fluorinated Fibres: PTFE, PVF, PVDF, and FEP. Poly(ether ether ketones): PEEK. Poly(phenylenesulphide): PPS. Poly(ether imide): PEI, PBI, and PBO. Unit VI - Technological Developments in the Manufacturing of Bicomponent Fibres, Importance and Applications of Bicomponent Fibres.				
Reference Books 1. High Performance Fibres, Edited by J. W. S. Hearle, Published by wood head publishing Ltd., England in association with Textile Institute Manchester 2. Carbon fibers by J. P. Donnet and R. C. Bansal, Marcel Dekker, New York 3. Hand book of Fibres Science and Technology, High Technology Fibres, Edited by Manachem Lewin and Jack Preston. 4. New fibers. T. Hongu and G. O. Phillips Ellis Horwood Ltd, Chichester, 5. Kevlar aramid fiber. By H.H. Yang. John Wiley and Sons, Chichester, New York, 6. Mukhopadhyay S. K., "Advances in Fibre Science" The Textile Institute. 1992, ISBN:1870812379 7. Gupta V.B. Textile Fibres: Developments and Innovations. Vol. 2, Progress in Textiles: Science and Technology. Edited by V.K. Kothari, IAFL Publications, 2000.				
Practical: Practicals are to be conducted based on the above Course Contents.				

for Jaykhani

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-507	Program Elective-II Textile Project Planning & Implementation	03	03	00
Course Contents Objectives: - To explain the logic of project concept and its development cycle - To describe in detail the technical analysis for raw material and utilities - To illustrate the correlation of money with time with examples - To explain requirements for appraisal and project implementation				
Course Contents Outcomes: At the end of the Course Contents students will be able to - Describe the logic of Project development cycle & identification of Investment - Explain the Basics of Technical Analysis for Material inputs & utilities - Explain the Time value of money with numerical examples - Discuss the study on Appraisal criteria & Project implementation steps				
Course Contents: - Unit I - Overview. Project development cycle, Objectives of investment, decision-making, Risk & return Identification of investment opportunities – Governmental regulatory framework – Generation & screening of project ideas – Project Identifications for an existing company. Unit II - Market & demand analysis – Information required for market & demand analysis – demand forecasting methods – market planning. Cost of Capital – Basic concepts – Cost of debt – cost of preference capital – cost of Capital – Weighted average cost of capital –Marginal cost of capital–Cost of capital for a new company. Unit III - Technical Analysis – Material inputs & utilities – Manufacturing process. Technology – Plant capacity – location & site – structures & civil works –Machineries & equipment's – Project charts & layouts – Work schedule –Need for tendering alternatives. Unit IV - Financial Analysis – Cost of Project – Means of finance – Estimation of Sales & Production – Cost of production – Working capital requirement& financing – Profitability projections – Breakeven point – Project cost flow statements – Projected balance sheet – Multi – year projection Unit V - Time value of money – Future value of single amount, Future value of an annuity –Present value of single amount – Present value of an annuity. Analysis of Risk – Types & measurement of project risk – Analytical derivation or simple estimation – Sensitivity Analysis – Scenario analysis–Selection of a project–Risk analysis in practice Unit VI - Appraisal criteria – Urgency, Payback period – Accounting, Debt service coverage ratio, Rate of Return, Net present value – Internal rate of return – Annual capital charge – Investment appraisal in practice. Project implementation – Forms of project organization – Project planning – project control – Human aspects of project management –Prerequisites for successful project implementation				
References Books: - Textile Project Management by A. Ormerod, The Textile Institute Publication. - Goal Directed Project Management by E.S. Andersen, K.V. Grude & Tor Hang, Coopers& Cybranl Publication. - Project, Planning Analysis, Selection Implementation & Review by Prasanna Chandra, Tata McGraw Hill Publishing Co. Ltd., - Industrial Organisation & Engg. Economics T.R. Banga & S.C. Sharma, KhannaPublishers, Delhi.				
Practical: Practicals are to be conducted based on the above Course Contents.				

List of Course Contents for Program Elective-II (Any One)

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-508	Program Elective-II Industrial Engineering and Management	03	03	00
Course Contents Objectives: 1. To explain need of industrial engineering and principles of management 2. To understand theoretical concepts and practical aspects of work studies. 3. To learn theoretical concepts ergonomics. 4. To understand theories of TQM.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Understand importance of industrial engineering and principles of management 2. Understand theoretical concepts and apply work studies. 3. Understand and apply theories of TQM. 4. Evaluate and design Work system design ergonomically.				
Course Contents:- Unit I -Introduction to Work Study: Definition: Purpose of study, objectives, brief history and evolution, work study and productivity, human factor in application of work study, scope, applications, relationship, between Productivity & standard of living, basic work content, excess work content Management, techniques to reduce excess work content due to product process and ineffective time in control of workers and Management. Unit II - Ergonomics: Introduction, Principles, Work system design, Man-machine system, Human behaviour and equipment design, Tools, Techniques and applications, Effect of environment on performance of worker, working conditions, prevention accidents and hazards, lighting, ventilation etc. Unit III - Method Study: Definition, Concept, Objectives and Procedure of method study, Flow and handling of materials; Process chart symbols, recording techniques like Flow process charts, Operation, Flow and Two handed Process charts, Flow diagram, String diagram, Multiple Activity chart, travel chart, Operation Analysis, Analysis of motion, analysis and critical examination of existing methods and development of improved methods, Motion economy, Design of work place layout, Therbligs, SIMO chart. Unit IV - Work Measurement: Definition, significance of work measurement; origin, development and procedure of work measurement, introduction to various work measurement techniques. Time Study and Other Works Measurement Techniques: Time study: definition, equipment for basic time study, time study forms and other equipment. Steps in use of techniques of time study; selecting the job, breaking the job into elements, approach to the worker, the elements, timing each element, Maynard Operation Sequencing Technique (MOST), Average and qualified worker, rating procedures, criteria affecting the choice of rating procedures, criteria affecting the choice of rating procedures, continuous timing, fly back timing, accumulative timing; standard ratings. Unit V - Comparison of observed and standard ratings, factors affecting the rate of working, scales of rating, rating factors, recording the rating, summarizing the study, allowances, calculation and application of allowances. Work sampling and production studies; General study of standard data & PTS. Unit VI - Quality management: Quality as a Corporate Strategy, What is Quality?, New Quality Concepts, Quality Circles, Kaizen, Contributions of Quality Management scientists- Dr. Deming, Crosby, Dr. Armrod V. Feigen Baum, Dr. Ishikawa- Seven tools of quality management. Total Quality Management, Roadmap for TQM, Implementation of TQM, Six Sigma, Five 'S', TIMWOOD 7-Seven Wastes, ERP etc.				

for PAXhale

Reference Books:

1. Introduction to work study - ILO
2. Motion & Time study Design & Measurement of Work - Ralph Barnes (Wiley Eastern).
3. Work Study - R.M. Currie & Faraday. (ELBS Pitman).
4. Hand Book of Industrial Engineering - Irson & Grant.
5. Productivity management - Concepts & Techniques- S.C. Sawhney.
6. Production and Operations Management – IInd Edition by S. N. Chary, published by Tata McGraw Hill
7. Testing and quality management by DR. V. K. Kothari

Practical:

Practicals and assignments are to be conducted based on the above Course Contents.

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-509	Program Elective-II Industrial Law and Economics	03	03	00
Course Contents Objectives: 1. To understand principles of economics and entrepreneurship 2. To know labor legislations 3. To apply statistical methods for data analysis. 4. To learn and apply DEA model for productivity measurement. 5. To do SWOT analysis of Indian Textile Industry				
Course Contents Outcomes: 1. Remember principles of economics and entrepreneurship 2. Knowledge of labor legislations 3. Use statistical methods for data analysis. 4. Apply DEA model for productivity measurement. 5. SWOT analysis of Indian Textile Industry				
Course Contents Unit I Basic concepts of micro and macroeconomics: Supply and demand, demand schedule and demand curve, demand function, supply schedule and supply curve, production function, factors of production, returns to scale, economies of scale, balance sheet of the company Unit II Basic econometrics: Definition, methodology of econometrics, regression analysis, regression versus correlation, nature and sources of data for econometric analysis, two variable regression analysis, use of EXCEL and SPSS for regression analysis Unit III Efficiency and productivity measurement: Basic concepts of efficiency measurement, selection of input and output variables, mathematical aspects of Data Envelopment Analysis, application of computer based softwares such as DEAP, EMS, productivity measurement approaches, partial factor productivity with its limitations, total factor productivity, productivity measurement using DEA software Unit IV Structure of Indian textile industry: Functions & organizational set-up of cotton and man-made industry, jute, silk, wool and woolen textiles industry, decentralized power loom sector, handlooms, handicrafts, public sector undertakings, textile research associations, government textile policies, export/import trends, domestic market, SWOT analysis of Indian textile industry, globalization and effect of WTO on Indian textile industry Unit V Industrial relations & Labour laws: Definitions & objectives of industrial relation, charge procedure, punishment & appeal, collective bargaining, employee grievances -nature, grievance handling procedure, workers participation in management, Industrial Disputes Act 1947, The Factories Act 1948, The Wages Act 1956, The Minimum Wages Act 1948, Workmen's Compensation Act 1923, The Gratuity Act 1972				

for Prof. Khule

Unit VI Entrepreneurship development:

Introduction, entrepreneur, modern concepts and qualities required, classification of entrepreneur, factors conducive for promoting entrepreneurship, EDC/EDP–training and development of women entrepreneur

Reference Books:

1. Principles of Economics by Alfred Marshall
2. Principles of Economics by H. S. Agrawal
3. An Introduction to Data Envelopment Analysis by R. Ramanathan
4. Basic Econometrics by Damodar N. Gujarati
5. Annual report of Ministry of Textiles, www.textilemin.nic.in
6. Compendium of Textile Industry by Textile Commissioner
7. Labour & Industrial Law by S. K. Mishra
8. Industrial & Business Management by Martand T. Telsang

Practical:

Case Studies and assignments are to be conducted based on the above Course Contents.

**First Year M. Tech.
Semester-II**

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PCC-TT-510	Modern Weaving Technology	04	03	02
Course Contents Objectives: 1. To explain technological significance in the design development of automatic winding machine and process parameters involved in it. 2. To explain technological significance in the design development of beam and sectional warping machine and process parameters involved in it. 3. To explain technological significance in the design development of sizing machine. 4. To explain technological significance in the design development of unconventional weaving machines machine and process parameters involved in it				
Course Contents Outcomes: At the end of the Course, students will be able to: 1. Understand technological significance in the design development of automatic winding machine and process parameters involved in it. 2. Critically analyze technological significance in the design development of beam and sectional warping machine and process parameters involved in it. 3. Critically analyze technological significance in the design development of sizing machine. 4. Evaluate design, functional, and constructional aspects of unconventional weaving machines.				
Course Contents Unit I - Need of modern developments in weaving and its benefits Modern developments in Winding: Introduction: Yarn count monitoring, online classification of yarn defects, classification of foreign fibres. Joining of yarns, splicing: Pneumatic splicing, Electrostatic splicing, Mechanical splicing. Precision winder, auto speed control, automation, spindle identification, waste control, environment controls. Uniform build of yarn package: Theory & practice. Effect of winding faults on H.S. warping and weaving. Unit II - Modern Warping and Sizing: Concept of modern warping, developments in creel, head stock and different control systems. Concept of modern sizing, developments in sow box, driving of cylinders and head stock, Development in sizing ingredients and different control systems. Unit II - Modern Looms: Suitability of H.S. weaving machines. Technical aspects, economic aspects, manufacturer and machine related criteria, growth of technology, economics. Air-Jet Looms: Principles of weft insertion, weft tensioning, weft length measurement, sley movement, take-up motion. Equation of motion for weft yarn. Air-jet velocity profile, main nozzle, air guides, relay nozzle, weft travel position in air-jet, energy requirements. Method of operation of main jet. The air stream in the main nozzle area and its action on the weft thread. Improvement jet system. Study on pneumatic weft insertion behavior in main nozzle. Interaction between air, yarn and guide system to increase the yarn velocity. Advancement of Air-jet looms. Range of applications. Unit III - Rapier weaving: Recent developments, development in filling insertion, picking speed, secure filling transfer, and versatility of Rapier of Rapier gripper. Free flight rapier, light rapier head, electronic filling tension controller, pre winder switch-off monitoring, filling detection at the end of insertion, pick finding, rapier cleaning devices. Electronic controls, developments in let-off mechanism and take-up mechanism, selvages. Efficiency of rapier loom.				

Dr. Prithula

Unit IV - Gripper Projectile Loom: Ballistic weft insertion, velocity and acceleration of projectile, work done in picking. Different kinds of projectiles uses, colour change at thread transfer. Some features of modern Sulzer looms. PU & PS series machines, Sulzer P7100. Economics of projectile weaving machine. Range of applications.

Unit V -Techno Economics of High Speed weaving: Introduction, parameters needed for techno economics analysis. Selection of weaving machines. Cost-width relationship, colour& weft mixing, noise level, yarn quality, floor & building. Economic analysis. Capital cost, accommodation cost, spare parts cost, energy cost, accommodation costs, and conclusion.

Unit VI - The position of cloth fell under stable weaving conditions: The function of take-up motion, the physical aspect of the cloth fell equation, the relation between beat-up force & pick spacing (the inverse distance equation). Bumping conditions, cloth fell equation under bumping condition.

Reference Books:

1. Principles of Weaving- Mark & Robinson.
2. Shuttleless Wvg. Machines- Oldrich Talavasek and Vladimir Svaty.
3. Weaving machines, mechanisms management – M.K. Talukdar,
4. High speed weaving-K. Jaychandran at all-PSG College of Technology, Coimbatore
5. SIZING-Materials, Methods, Machines- Ajgaonkar, Talukdar and Wadekar

Practical:

Practicals are to be conducted based on the above Course Contents.

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PCC-TT-511	Evaluation Techniques of Textiles	04	03	02
Course Contents objectives: 1. To learn basic science of testing of fibre, yarn and fabric. 2. Understand quality measurement systems. 3. Learn the functional testing methods. 4. Study in deep the Spectrometer and its application in textile measuring techniques.				
Course Contents outcome: 1. Apply knowledge of science & engineering for textile testing 2. Analyze textile material using various test methods 3. Learning new methods/technology for testing Understand textile testing from environmental angle Writing textile testing reports 4. Assess textile material in relation to health and safety				
Course Contents Unit I - Textile Quality Management Overview: Quality and its importance, ISO system Unit II - Fiber Testing: AFIS- Principle, working and applications, Advantages over HVI Yarn Testing: CTT- Principle, working and applications Unit III - Fabric Testing: KES- Principle, working and applications; FAST- Principle, working and applications Unit IV - Functional Testing: Tests for Thermal Transmission, Moisture Vapour Transport, Combined Effect of Thermal and Moisture properties, Electrical Properties, Flammability Unit V - IR Spectroscopy: Principle and its applications for Cotton, Blend Analysis Testing for UV Protection: In vitro and In vivo Unit VI - Labelling: Woolmark, Care, etc.				
References: 1. Progress in Textiles: Science & Technology, Volume I, Testing and Quality Measurement, V K Kothari, IAFL publications, India, 1999 2. Intelligent Textiles and Clothing, H r Mattila, Woodhead Publishing Limited, 2006 3. Textiles for Protection, R A Scott, Woodhead Publishing Limited, 2005 4. Textile Testing Fiber, Yarn & Fabric, Arindam Basu, SITRA, 2001 5. Smart Textiles for Medicine and Healthcare, L Van Langenhove, Woodhead Publishing Limited, 2007 6. www.nptel.ac.in				
Practical: Practicals are to be conducted based on the above Course Contents.				

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List of Course Contents for Program Elective-III -(Any One)

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-512	Program Elective-III Technical Textile - I	04	03	02
Course Contents Objectives: 1. To introduce technical yarns and technical textiles. 2. To describe Coating & Lamination Textiles and application of technical textiles. 3. To explain different finishes on yarn. 4. To describe Miscellaneous Applications.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Describe introduction of technical textiles 2. Explain Coating & Lamination Textiles and application of technical textiles. 3. Describe different Novel finishes on yarn. 4. Describe Shape memory polymer and its outcomes 5. Explain Miscellaneous Applications.				
Course Contents :- Unit I - Introduction, types of technical yarn, yarn characteristics: monofilament, multifilament, intermingled yarn, tape yarn, core spun yarn-Non elastic core and elastic core, plied/ folded yarn, cabled yarn, braided yarn. Unit II -Tape yarn production technique, fibrillated tape yarns. Filament wrapped yarns. Properties and performance of technical yarns. Unit III -Textile Coating & Laminating: Coating materials, coating polymers such as PVC, PVDC, PU, PTFE, Acrylic polymers, rubber and its derivatives, EVA, CARBON NANO TUBES. Unit IV -Principles of coating, aqueous coating, hot melting coating, metal coating, plaiting, plasma treatment. Different methods of coating. Unit V -Novel technical textile yarns: Introduction, reflective yarns, classification, and manufacturing process. UV protected yarn, preparation of UV protection yarn, metallic and metalloplastic yarns, and manufacturing techniques. Antimicrobial yarns, treatments, durable antimicrobial fibre/yarn. Yarns for specific purpose. Anti-static yarn, anti-stress yarn, anti-allergic yarn, soluble yarn. Unit VI -Electro conductive yarns, manufacturing process, Measurements, EMSE, applications High modulus yarns such as Glass, Carbon, and Ceramic, Basalt fibers, HPPE fibres, their properties and uses. Shape memory polymer yarns, manufacturing process, applications. Plasma treated yarns for biomedical applications. Chemistry of plasma processing. Industrial sewing threads, structure of sewing threads, thread finishing, yarn numbering. Threads for very high temperature. Surgical threads, classifications, manufacturing process, characterization				
References Books: 1. A. R. Horrocks & S. C. Anand, Handbook of technical textiles, The Textile Institute, Wood Head Publication Ltd., 2007 2. R. Alagiruswami and A. Das, Technical Textile yarns, The Textile Institute, Wood Head Publication Ltd., 2010 3. Wellington Sears Handbook of Industrial Textiles by Sabit Adanur, Technomic Publishing Co. INC, Lancaster, Basel, 1995 4. High performance fibres by P. Bajaj and A. K. Sengupta, The Textile Institute 5. Industrial Applications of Textiles: Textiles for Filtration and Coated Fabrics, P. Bajaj and A. K. Sengupta, The Textile Institute, Textile Progress, Vol 14, No.1				
Practical: Practicals are to be conducted based on the above Course Contents.				

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-513	Program Elective-III Advanced Textile Wet Processing:	04	03	02
Course Contents Objectives: 1. To educate basic knowledge of procedures, machinery & chemistry of different dyeing processes. 2. To educate basic developments of dyeing machines. 3. To know the colour measurement techniques. 4. To understand the modern printing, advanced finishing and eco treatments.				
Course Contents Outcomes: 1. Student developed understanding of engineering knowledge in different dyeing and finishing operation & technology for textile fabrics. 2. Student developed understanding of problem analysis skills in identifying different types of dyes and finishes applied in textile & their quality evaluation. 3. Student developed understanding of modern tools & machines required for different dyeing, printing and finishes application in fabrics. 4. Students developed understanding to work as individual & team work through conducting different individual & group assignment & practical job work. 5. Students developed understanding in effective communication through repeated written and oral test & assignment of the subject				
Course Contents Unit I - Dyeing Process Developments & Kinetics: Influence of fibre structure on dye uptake; Kinetics of Dyeing; Developments in dyes and dyeing processes for the dyeing of various textile substrates with various dye classes. Advances in Mass coloration. Unit II - Dyeing Machines Developments: Advances in cheese dyeing machines. Advances in Beam dyeing; Advances in soft flow dyeing machines, Advances in jet dyeing machines. Developments in jiggers, Continuous dyeing machineries & its developments, Various dyeing defects caused by the above machineries; Garment Dyeing & it's modern machineries. Unit III - Color Measurements: Spectrophotometric color measurement & analysis of dye solutions; Fastness properties of dyed materials and their assessment. Identification of dyes on fibres. Unit IV - Pollution & Eco Treatments: Pollution aspects of textile dyeing; Modern approaches to eco-friendly wet processing of woven and knitted textiles. Eco-friendly dyes and their method of dyeing; Technology and principles of effluent treatment. Advanced color removal technologies, recovery and reuse of water and chemicals. Unit V - Modern Printing: Principle and working of fully automatic flatbed screen printing machine ;Rotary Printing machine; Transfer Printing machine-Digital Printing; Garment Printing machines; Various practical problems & possible remedies in the above printing machineries; Modern developments in textile printing machines/techniques. Unit VI - Modern Finishing: General overview of the recent technological developments in the area of textile finishing machines/techniques. Formaldehyde free finishes; stone wash, Enzyme wash, Bio – polishing, Acid wash, sand blasting, leather finish, rubbery touch, feather touch, easy care finishes, wrinkle free and wrinkle resistant finish, water repellent finish, UV protective garments, Anti – microbial/ anti – bacterial inhibition finish, silicone softeners, fire retardant finishes for garments, functional finishes for garments; Micro- encapsulation and its relevance in textile finishing application.				

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Reference Books:

1. Dyeing & Chemical Technology of Textile Fibres, By: **E. R. Trotman**; Published By: Charles Griffin & Company Ltd.
2. Fundamentals & Practices in Coloration of Textiles, By: **Chakraborty J. N.**; Published By: Woodhead Publishing India in Textiles.
3. Technology of Textile Printing; By- **R.S.Prayag**; Published By: Mrs. R.S.Prayag, Dharwad, and Karnataka State.
4. Textile Printing; Edited By Leslie W.C. Miles, Published By: Society of Dyers & Colourists(2003)
5. Technology of Textile Processing – Vol-IV "Technology of Printing" By: **Dr.V.A.Shenai**; Published By: Sevak Publications, Mumbai.
6. Technology of Textile Processing - Vol.6, "Technology of Dyeing" By: **Dr. V. A. Shenai**; Published By: Sevak Publications, Mumbai.
7. Technology of Textile Processing - Vol.2, "Chemistry of dyes & Principle of dyeing" By: **Dr. V. A. Shenai** ; Published By: Sevak Publications, Mumbai.
8. Book of Papers: **Convention on Natural Dyes-Dec-1999**; Published By: IITD.
9. Schindler W.D. And Hauser P.J., "Chemical Finishing of Textiles", Woodhead Publishing Limited, Cambridge, 2004.
10. Chemistry & Technology of Fabric Preparation & Finishing, By: Dr. Charles Tomasino, Published By: Dept. Of Textile Engg, Chemistry & Science, College of Textiles, North Carolina State University, North Carolina.
11. Textile Finishing; By: **R.S Prayag**; Published By: Mrs. L.R.Prayag, Dharwad, Karnataka State.

Practical:

Practicals are to be conducted based on the above Course Contents.

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-514	Program Elective-III Garment Technology & Merchandising:	04	03	02
Course Contents Educational Objectives (CEO): The student should learn- 1. Apply basic elements and principles of garment manufacturing. 2. Learn garment manufacturing process 3. Design and produce a buyer specification garment 4. Elucidate the management of the Garment business 5. Illustrate the fashion marketing and merchandising process. 6. Study Retailing and Global Sourcing Strategies, Supply Chain and demand chain analysis				
Course Contents Outcomes (COs): Upon successful completion of this Course Contents, the student will be able to: 1. Remember basic elements and principles of garment manufacturing. 2. Handle and control garment manufacturing process 3. Create a Design and produce a buyer specification garment 4. Evaluate the management of the Garment business 5. Apply the fashion marketing and merchandising process. 6. Depict Retailing and Global Sourcing Strategies, Supply Chain and demand chain analysis				
Course Contents Unit I - Introduction – Sectors and structure of apparel industry, Overview of Indian garment industry, Nature & scope of apparel manufacturing industry and its developments in recent years. Unit II - Sequence of garment manufacturing process. Latest developments in garment technology. Grading, Types and making of lay plan, Requirements of spreading, types and methods of spreading, spreading equipment's and tools, cutting equipment's and tools and their modernization, size charts etc. Unit III - Technology of Sewing machines, Sewing defects. Fabric sew ability, principles of selecting proper stitch and seam types, Effect of stitch type on elasticity and strength, Effect of stitch type on seam slippage. Stitch less garments. Garment dyeing and finishing, Value added garments. Pressing & finishing: object, classifications, means, components, machinery and equipment's, garment finishing and inspection, Quality Standards of some giant retailers, TUV, SGS and ASTM testing standards. Unit IV - Production technology: Manual systems make through systems, straight line systems, modular production systems, unit production systems, quick response systems. Ware housing: Handling equipment's, storage equipment's, packing equipment's. Application of CAD/CAM in garment manufacturing Unit V - Definition and role of merchandiser, Buying indication, Market structure in export markets. Design buying influences. Merchandising and Range Development: Fashion forecast for Europe and USA, elements of design, Interpretation of forecast for brands incolor, silhouette and fabrics. Unit VI - General range development and specific buyer range developments. Colors: International interpretation of colors. Market Research: Buyer behavior, apparel brand				

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management, advertising & promotion, Retail management, Pricing, Approaches for global markets: Effect on Global Economy Analysis of effect of international production and distribution of textiles and apparel goods on the global economy, Cost-plus and backward pricing, Elements of cost plus approach & cost sheet, Merchandise line pricing.

Reference Books:

1. The Technology of clothing manufacturing by Harold Carr, Barbara Latham, Publisher – Blackwell scientific publications
2. Handbook for designing by Ritu Jindal, S. Malhan, Publisher - Mittal Publications
3. Managing Productivity in the apparel Industry by Rajesh Bheda, Michael T. Fralix, Publisher – CBS publications & distributors
4. The apparel industry in India, Ila Kantilal
5. CAD/CAM in clothing & Textiles by Grey and Stephens Publications – Gower Publishing, Hampshire, 1998.
6. Fashion Buying, Elaine Stone
7. Principles of Fashion Merchandising, Sidney Packard

Practical:

Practicals are to be conducted based on the above Course Contents.

List of Course Contents for Program Elective-IV - (Any One)

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-515	Program Elective-IV Knitting & Nonwoven Technology	04	03	02
Course Contents Objectives: 1. This Course Contents work will help the student to enhance their knowledge in knitting & nonwovens process. 2. Also give the student an independent knowledge of machine & structure about knitting & nonwoven. 3. To learn about variables, testing, defects in process and products. 4. To study techno economics, uses & applications 5. To study calculations involved in knitting & nonwovens.				
Course Contents Outcome: 1. It will update the knowledge about modern aspects of warp knitting and non-woven production and application. 2. Students will gain detailed knowledge of machine & structure. 3. Students will be able to set the process and testing of products produced. 4. Student will come to know techno economics, uses & applications 5. Students will evaluate and analyze calculations involved in knitting & nonwovens.				
Course Contents Unit I -Warp knit fabrics; warp knit v/s woven construction; warp knits vs weft knit. Machine required for single jersey, rib, pearl and interlock double jersey structures. Structure, fabric defects, latest developments & calculation. Unit II -Classes of warp knitting machinery, knitting cycle, Tricot, Raschel machines. Knitting elements 7 cycles of Tricot and Raschel machine. Unit III -Plain Tricot structures, knitted with two full set guide bars, two bar Tricot, Shark skin, Queens cord, Velour and Velvet structures, Satin, overfed pile structures, reverse lock knit. Differences between Tricot and Rachel machines and fabrics. Yarns for warp knitting: -Materials for warp knitting, filament and spun yarns, unconventional yarns, important yarn properties for warp knitting, winding and warping for warp knitting. Faults in warp knits. Warp knitting calculations Unit IV -Classification & process of nonwovens, web forming, laying and bonding different techniques Unit V -Melt spun and melts blown nonwovens, process, variables, defects & applications. Unit VI -Nonwoven fabric finishing: types of finishing techniques, Fabric inspection, Testing & study of process variables, advantages, disadvantages and techno economics of all above non-woven technologies.				
Reference Books: 1. Fundamentals & advances in Knitting Technology- S C Ray, WPI 2. Knitting Technology by David J. Spencer, WPI 3. Knitting Technology by D.B. Ajgoankar, Universal Publishing Corporation, Mumbai. 4. Nonwoven Process Performance & Testing –Turbak 5. Nonwoven Fabric Construction Synthetic Fibres – Jan-Mar 2007. 6. Proceedings of the Seminar- Nonwoven Technology Market & Product Potential, IIT, New Delhi Dec. 2006. 7. Handbook of nonwovens, Edited by S.J.Russell, Woodhead Publishing, CRC Press, Washington DC, 2007 8. W.Albrecht, H. Fuchs and W.Kettelmann, Nonwoven Fabrics: Raw Materials, Manufacture, Applications, Characteristics, Testing Process, Wiley-VCH, Verlag GmbH & Co.KGaA, Weinheim, 2003. 9. M.S. Casper, Nonwoven Textiles, Noyes Data Corp.(Park Ridge, N.J), 1975. 10. M. McDonald, Nonwoven Fabric Technology, Park Ridge, NJ: Noyes Data, 1971				
Practical: Practicals are to be conducted based on the above Course Contents.				

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Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-516	Program Elective-IV Textile Composites	04	03	02
Course Contents Objectives 1. To explain requirements of fibre and matrix for composite fabrication & their types 2. To describe the fibre-matrix interactions in unidirectional lamina 3. To explain details of various methods of composite fabrication 4. To explain properties of composites and their applications				
Course Contents Outcomes At the end of the Course Contents students will be able to 1. Understand the logic, need, requirements of composites based on end use 2. Elucidate the manufacturing of the composites and fibre used for fabrication 3. Evaluate the performance of composites including fibre matrix interactions 4. Discuss the 3D textile structural composites				
Course Contents 1. Composites: Definition, Objectives, Classification, Applications 2. Matrix Materials: Polymers used, Properties of polymers, Thermoset and thermoplastic resins, Nonpolymeric materials 3. Fabrication: Hand lay, Bag molding, Pultrusion, Blow molding, Preformed molding, etc. 4. Mechanics: Isostress, isostrain conditions, Critical Fiber Length, Critical Fiber volume, Calculations for stress, strain and modulus; changes for continuous to discontinuous fibers, Failure mechanism 5. Applications: For structural engineering, electrical, civil, aerospace, defense, automobile, sports, medicine and others 6. Surface treatments, Flammability and fire resistance of composites, Laminated composite				
Reference Books: 1. Design and Manufacture of Textile Composites, Long C A, Publisher Woodhead Publishing Series in Textiles 2. Composite Materials, K Srinivasan, Publisher Narosa Publishing House, Delhi 3. Analysis and Performance of Fiber Composites, Bhagwan D. Agarwal, Lawrence J, Broutman, and K Chandrashekhara 4. Composite Materials: Design and Applications, Daniel Gay, Suong V Hoa and Stephen W Tsai 5. Composite Material Science & Engineering, Springer Verlag 6. 3D Textile Reinforcements in composite Materials, Antonio Miravete, Publisher Woodhead Publishing Series in Textiles 7. www.nptel.ac.in				

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PEC-TT-517	Program Elective-IV Technical Textile - II	04	03	02
Course Contents Objectives: 1. To develop core knowledge of technical textiles to serve industry. 2. Understand different applications of medical textiles, sports textiles, protective clothing and geotextiles 3. Study different design techniques of construction of technical textiles 4. Design methods for stabilization of soil 5. Create new design with all your knowledge.				
Course Contents Outcomes: 1. Fundamental of science of Technical Textiles. 2. The subject requires collection of information from journals, net and understands the complex problem deeply to design different protective clothing such as EMI protective clothing, radiation protective clothing and other mechanical protective clothing. 3. Design of different Technical Textiles such as flame-retardant fabrics, chemical protective clothing, sports textiles, medical textiles, geotextiles 4. Application of Technical Textile and safety of human life & society using mechanical protective clothing, bullet proof cloth for military uses etc. 5. Designing and analysis of medical textiles, sports textiles, protective clothing and geotextiles				
Course Contents Unit I -Medical textiles: Introduction, textile materials used-category wise, fibre used, wound care, bandages, extracorporeal devices, implantable materials, sutures, soft tissue implants, orthopaedic implants, cardiovascular implants, healthcare and hygiene products. Unit II - Sports footwear: Functional design of sportswear, functional fit of sport footwear, biomechanics of the foot, functional materials and components in sport footwear Unit III - Mechanical protective clothing: Introduction, materials used, gloves, chain saw clothing, electrostatic protection, Physical requirements for military textiles, military combat clothing systems, Bullet proof fabrics principles & designs, mechanics of ballistic impact, textile materials for ballistic protection, design of ballistic vests and helmets, ballistic testing and evaluation, Clean room fabric, radiation protection, Protective clothing for space. Unit IV - Chemical protective clothing Different types of protective materials, components, clothing materials, protection from liquid, toxic fumes and gases, Performance evaluation of chemical protective clothing, chemical, biological and radiation hazards and their prevention. Unit V - Heat and flame protection: Fire fighter's protective clothing, Military flame-retardant, heat protective textiles, military flame and heat threat, criteria for protection of individual, toxic fumes and smoke, thermoplastic melt hazard, flame retardant textiles in military use. Unit VI - Geo-textiles: An overview of Geosynthetics their uses in different fields. Testing of Geotextiles and their needs, Designing for reinforcement on roads. Introduction, uses of Geotextiles. Mode of action. Design method with geotextiles by Giroud and Noiry method. Construction with geotextile. Design with geogrids for payments, Erosion control, and conventional graded filter. Geotextile filter. Geotextile filter criteria. Geotextile survivability, construction criteria, Designing or stabilization with geotextile walls. Construction details, design method.				

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Reference Books:

1. Soil Mechanics-B.C.Punmia
2. Non-woven bonded fabrics- J.Lunenschlos, Publisher- Ellis Harwood Series.
3. Designing with Geosynthetics-Robert M.Koerner, Publisher-Prentice Halla, Englewood cliff
4. A. R. Horrocks& S. C. Anand, Handbook of technical textiles, The Textile Institute, Wood Head Publication Ltd., 2007
5. R. Alagiruswami and A. Das, Technical Textile yarns, The Textile Institute, Wood Head Publication Ltd., 2010
6. Handbook of Industrial Textiles by SabitAdanur, Technomic Publishing Co. INC, Wellington Sears Lancaster, Basel,1995
7. Industrial Applications of Textile by R. S. Goy & J. A. Jenkins, Textile progress , 1970, March Vol. II
No.1Textiles in sports by R. Shishoo, Wood Head Publication, The Textile Institute, 2005

Practical:

Practicals are to be conducted based on the above Course Contents.

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
PRJ-TT-518	Mini Project and Seminar	01	--	02

Course Contents Outcomes:

At the end of the Course Contents students will be able to

1. Describe the problem/idea and review and summarize the literature for the topic of the identified problem
2. Illustrate the suitable design of experiments including experimental plan.
3. Explain the concepts of design, development, construction, and fabrication of innovative product/system for the project title
4. Use various tools of testing and statistical analysis for the data in order to draw relevant conclusions.

Rationale:

The mini project will involve the design, development, construction, and fabrication of innovative product/system approved by the department. This is a laboratory-oriented Course Contents which will provide a platform to students to enhance their practical knowledge and skills. Each student must keep a project notebook. The notebooks will be checked periodically throughout the semester by the guide and also during the internal viva, as part of the project grade.

Guidelines:

1. Students should select a problem which addresses some textile industry problem, or other product developments in textiles. One mini project per semester per student
2. The selected topic for mini project should be based on development/fabrication of innovative product which he/she learnt during Course Contents work.
3. Students should understand testing of various instruments relating to topic of mini project.
4. Execution of mini project should be carried out by students only under guidance of allotted faculty. One faculty per student.
5. Students should develop a necessary product with product specifications with reference to end use.
6. Students should see that final product submitted by them is in working condition.
7. 15-20 pages report to be submitted by students in prescribed guidelines. Presentation is for 10 minutes.
 - Groups of students cannot be permitted to work on a single mini project. Individual student has to carry out the mini project.
 - A demonstration and internal oral examination on the mini project also should be done at the end of the semester.
 - Department may arrange demonstration with poster presentation of all mini projects developed by the students at the end of semester.
 - It is desirable that the product developed by the students have some novel features.
 - A test of significance should be applied to the test results to ascertain the conformity of significant difference

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**Second Year M. Tech
SEMESTER-III**

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
DIS-TT-601	Dissertation - I	18	00	36
Course Contents Objectives: 1. To identify the problem/idea and review and summarize the literature for the topic of the identified problem. 2. To describe the process flow for undertaking the research/survey trials with appropriate standards and process variables. 3. To design, development, construction, and fabrication of innovative product/system for the final submission. 4. To explain various tools of testing and statistical analysis for the data in order to draw relevant conclusions.				
Course Contents Outcomes: At the end of the Course Contents students will be able to 1. Describe the problem/idea and review and summarize the literature for the topic of the identified problem. 2. Illustrate the suitable design of experiments including experimental plan. 3. Explain the concepts of design, development, construction, and fabrication of innovative product/system for the project title. 4. Use various tools of testing and statistical analysis for the data in order to draw relevant conclusions.				
Rationale: The Dissertation work of Phase-I is mainly the 50% of the project work. This includes literature review, design of experiment, selection of proper method of experiment, starting of work.				
Guidelines for Dissertation Phase I: 1. Students should complete the literature review. 2. Students should design the experiment 3. Select the proper method of testing 4. Select the materials 5. start the work and get result 6. Mid - Project Evaluation will be done at the end of Semester.				

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
HSSM - 601	Indian Knowledge System	2	2	00
Course Contents Objectives: 1. Creating awareness amongst the youths about the true history and rich culture of the country. 2. Understanding the scientific value of the traditional knowledge of Bharata; 3. Promoting the youths to do research in the various fields of Bharatiya knowledge system. 4. Converting the Bharatiya wisdom into the applied aspect of the modern scientific paradigm. 5. Adding career, professional and business opportunities to the youths. 6. It is also believed that after completion of this course, the students will get a holistic insight into the understanding of the working of nature and life.				
Course Contents Outcomes: After completion of course, students will be able to CO1 Understand and appreciate the importance of ancient knowledge to a society. CO2 Familiarize with the key components of the IKS CO3 Develop some appreciation of IKS historicity				
Course Contents: Unit 1: Bharatiya Civilization and Development of Knowledge System (4hours) Genesis of the land, Antiquity of civilization, On the Trail of the Lost River, Discovery of the Saraswati River, the Saraswati Sindhu Civilization, Traditional Knowledge System, The Vedas, Main Schools of Philosophy (6+3), Ancient Education System, the Takshashila University, the Nalanda University, Alumni, Knowledge Export from Bharata. Unit2: Arts, Literature and Scholars (4hours) Art, Music, and Dance, Nataraja: A Masterpiece of Bharatiya Art, Literature, Life and works of Agastya, Lopamudra, Ghosa, Valmiki, Patanjali, Vedavyasa, Yajnavalkya, Gargi, Maitreyi, Bodhayana, Charaka, Susruta, Jivaka, Nagarjuna, Kanada, Patanjali, Kautilya, Panini, Thiruvalluvar, Aryabhata, Varahamihira, Adi Shankaracharya, Bhaskaracharya, Madhavacharya. Unit3: Science, Astronomy, and Mathematics(4hours) Concept of Matter, Life and Universe, Gravity, Sage Agastya's Model of Battery, Velocity of Light, Vimana: Aeronautics, Vedic Cosmology and Modern Concepts, Bharatiya Kalaganana, Kerala School for Mathematics and Astronomy, History and Culture of Astronomy, Sun, Earth, Moon, and Eclipses, Earth is Spherical and Rotation of Earth, Archaeoastronomy; Concepts of Zero and Pi, Number System, Pythagoras Theorem, and Vedic Mathematics. Unit4: Engineering, Technology and Architecture(4hours) Pre-Harappan and Sindhu Valley Civilization, Laboratory and Apparatus, Juices, Dyes, Paints and Cements, Glass and Pottery, Metallurgy, Engineering Science and Technology in the Vedic Age and Post-Vedic Records, Iron Pillar of Delhi, Rakhigarhi, Mehrgarh, Sindhu Valley Civilization, Marine Technology, and Bet-Dwarka. Unit5: Life, Environment and Health(4hours) Ethnic Studies, Life Science in Plants, Anatomy, Physiology, Agriculture, Ecology and Environment, Ayurveda, Integrated Approach to Health care, Medicine, Microbiology, Medicine, Surgery, and Yoga, etc.				
References: 1. Textbook on The Knowledge System of Bharata by Bhag Chand Chauhan. 2. History of Science in India Volume-I, Part-I, Part-II, Volume VIII, by Sibaji Raha, et al., National Academy of Sciences, India and The Ramkrishan Mission Institute of Culture, Kolkata (2014). 3. Pride of India- A Glimpse of India's Scientific Heritage edited by Pradeep Kohle et al. Samskrit Bharati (2006). 4. Vedic Physics by Keshav Dev Verma, Motilal Banars ideas Publishers (2012). 5. India's Glorious Scientific Tradition by Suresh Soni, Ocean Books Pvt.Ltd. (2010).				

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**Second Year M. Tech
SEMESTER-IV**

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
DIS-TT-602	Dissertation- II	18	00	36
Course Contents Objectives: 1. Continuation of problem given in third semester 2. Testing and evaluation of problem using proper statistical aids. 3. Validation of results, paper publishing. 4. To design, development, construction, and fabrication of innovative product/system for the final submission				
Course Contents Outcomes: Course Contents Outcomes: At the end of the Course Contents, students will be able to: 1. Describe the problem/idea and review and summarize the literature for the topic of the identified problem. 2. Illustrate the suitable design of experiments, including an experimental plan. 3. Explain the concepts of design, development, construction, and fabrication of innovative product/system for the project title. 4. Use various tools of testing and statistical analysis for the data in order to draw relevant conclusions				
Rationale: The Dissertation work of Phase II is mainly the completion of the remaining 50% of the project work. This includes the compilation of results, results and discussions, conclusions.				
Guidelines for Dissertation Phase II: 1. Students should complete and compile the trials, testing. 2. Students should propose a complete thesis writing with given guidelines 3. Students will be ready for the internal Viva with synopsis, objectives, plan of work, and results and discussion.. 4. The results and discussion will be as per in line with the plan of work. No deviation is allowed. 5. Project Evaluation will be done at the end of Semester.				

Code	Name of the Course Contents	Total credits	Lectures /week	Practical/ Week
CCC - 602	Yoga and Sports	0	0	2
Course Contents Objectives: 1. To make the students understand the importance of sound health and fitness principles as they relate to better health. 2. To expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness. 3. To create a safe, progressive, methodical, and efficient activity-based plan to enhance improvement and minimize the risk of injury. 4. To develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health				
Course Contents Outcomes: On successful completion of the course the students will be able CO1. To practice Physical activities and Hatha Yoga focusing on yoga for strength, flexibility, and relaxation. CO2. To learn techniques for increasing concentration and decreasing anxiety which leads to stronger academic performance. CO3. To learn breathing exercises and healthy fitness activities. CO4. To perform yoga movements in various combinations and forms. CO5. To assess current personal fitness levels. CO6. To demonstrate an understanding of sound nutritional practices as related to health and physical performance.				
Course Contents: Unit 1: Introduction to Physical Education (1 hour) Meaning and definition of Physical Education Aims and objectives of Physical Education Changing trends in Physical Education Unit 2: Olympic Movement (2 hours) Ancient and Modern Olympics (Summer and Winter) Olympic Symbols, Ideals, Objectives, and Values Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhyan Chand Award, Rajiv Gandhi Khel Ratna Award, etc.) Unit 3: Physical Fitness, Wellness & Lifestyle (2 hours) Meaning and Importance of Physical Fitness & Wellness Components of Physical fitness Components of Health-related fitness Components of wellness Preventing Health Threats through Lifestyle Change Concept of Positive Lifestyle Unit 4: Fundamentals of Anatomy & Physiology in Physical Education, Sports and Yoga (2 hours) Define Anatomy, Physiology & Its Importance Effect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System, etc.) Unit 5: Kinesiology, Biomechanics & Sports (2 hours) Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports Newton's Law of Motion & its application in sports Friction and its effects in Sports Unit 6: Postures (1 hour) Meaning and Concept of Postures. Causes of Bad Posture. Advantages & disadvantages of weight training.				

for 

Concept & advantages of Correct Posture.

Common Postural Deformities - Knock Knee, Flat Foot, Round Shoulders, Lordosis, Kyphosis, Bow Legs, and Scoliosis.

Corrective Measures for Postural Deformities

Unit 7: Yoga (1 hour)

Meaning and Importance of Yoga

Elements of Yoga

Introduction - Asanas, Pranayama, Meditation & Yogic Kriyas

Yoga for concentration and related Asanas (Sukhasana, Tadasana, Padmasana, Shashankasana)

Relaxation Techniques for improving concentration - Yog-nidra

Unit 8: Yoga and Lifestyle (2 hours)

Asanas as preventive measures.

Hypertension: Tadasana, Vajrasana, PavanMuktasana, ArdhaChakrasana, Bhujangasana, Shavasana.

Obesity: Procedure, Benefits, and contraindications for Vajrasana, Hastasana, Trikonasana, Ardha Matsyendrasana.

Back Pain: Tadasana, Ardha Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana.

Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, PavanMuktasana, Ardha Matsyendrasana.

Asthma: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.

Unit 9: Training and Planning in Sports (1 hour)

Meaning of Training

Warming up and limbering down

Skill, Technique & Style Meaning and Objectives of Planning.

Tournament - Knock-Out, League/Round Robin and Combination.

Unit 10: Psychology & Sports (1 hour)

Definition & Importance of Psychology in Physical Edu. & Sports

Define & Differentiate Between Growth & Development

Adolescent Problems & Their Management

Emotion: Concept, Type & Controlling of emotions

Meaning, Concept & Types of Aggressions in Sports

Psychological benefits of exercise.

Anxiety & Fear and its effects on Sports Performance.

Motivation, its type & techniques. Understanding Stress & Coping Strategies.

Unit 11: Doping (1 hour)

Meaning and Concept of Doping

Prohibited Substances & Methods

Side Effects of Prohibited Substances

Unit 12: Sports Medicine (1 hour)

First Aid - Definition, Aims & Objectives. Sports injuries: Classification, Causes & Prevention.

Management of Injuries: Soft Tissue Injuries and Bone & Joint Injuries

Unit 13: Sports/Games

(Following subtopics related to any one Game/Sport of choice of student out of: Athletics, Badminton, Basketball, Chess, Cricket, Kabaddi, Lawn Tennis, Swimming, Table Tennis, Volleyball, Yoga, etc.)

History of the Game/Sport. Latest General Rules of the Game/Sport. Specifications of Play Fields and Related Sports Equipment. Important Tournaments and Venues. Sports Personalities. Proper Sports Gear and its Importance

References:

1. Modern Trends and Physical Education by Prof. Ajmer Singh.
2. Light on Yoga By B.K.S. Iyengar.
3. Health and Physical Education - NCERT (11th and 12th Classes)

List of Experiments:

Students have to practice different Asanas mentioned in the syllabus, study different sports

for *Prithvi*